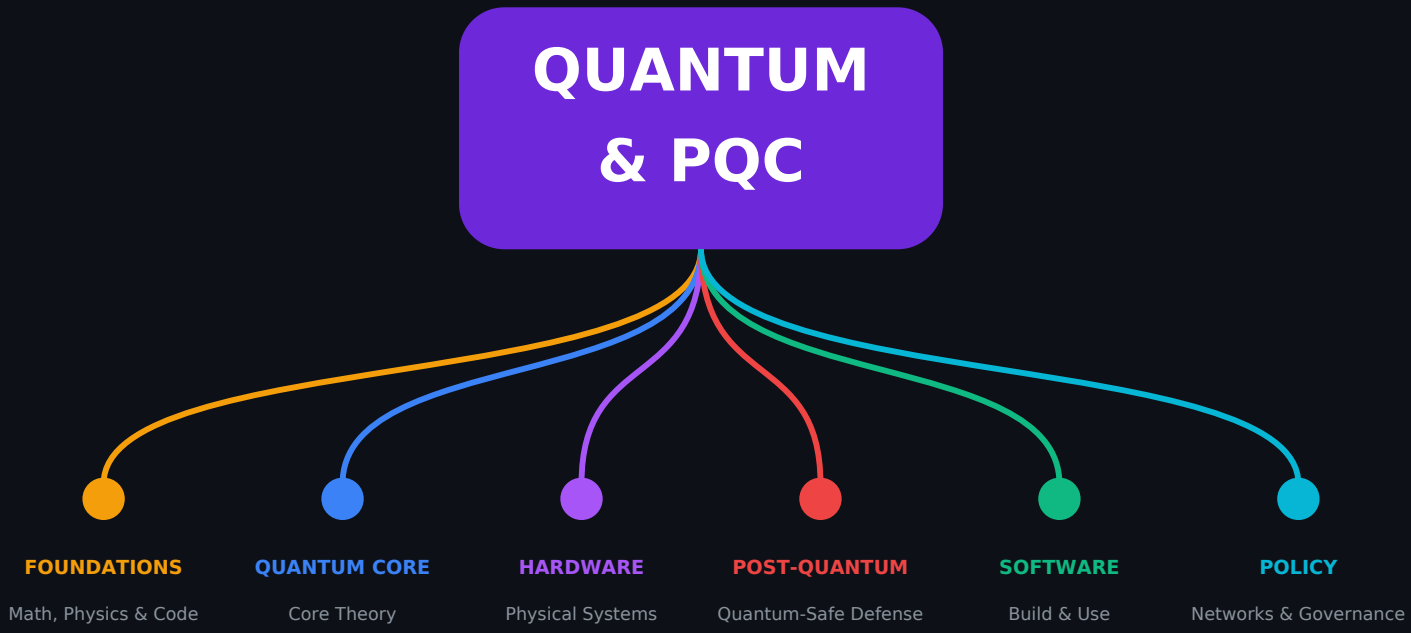
The Quantum Computing & PQC Tree

Quantum Computing | Hardware | Post-Quantum Cryptography | Software | Policy



6	29	113	40
branches	fields	tools	links

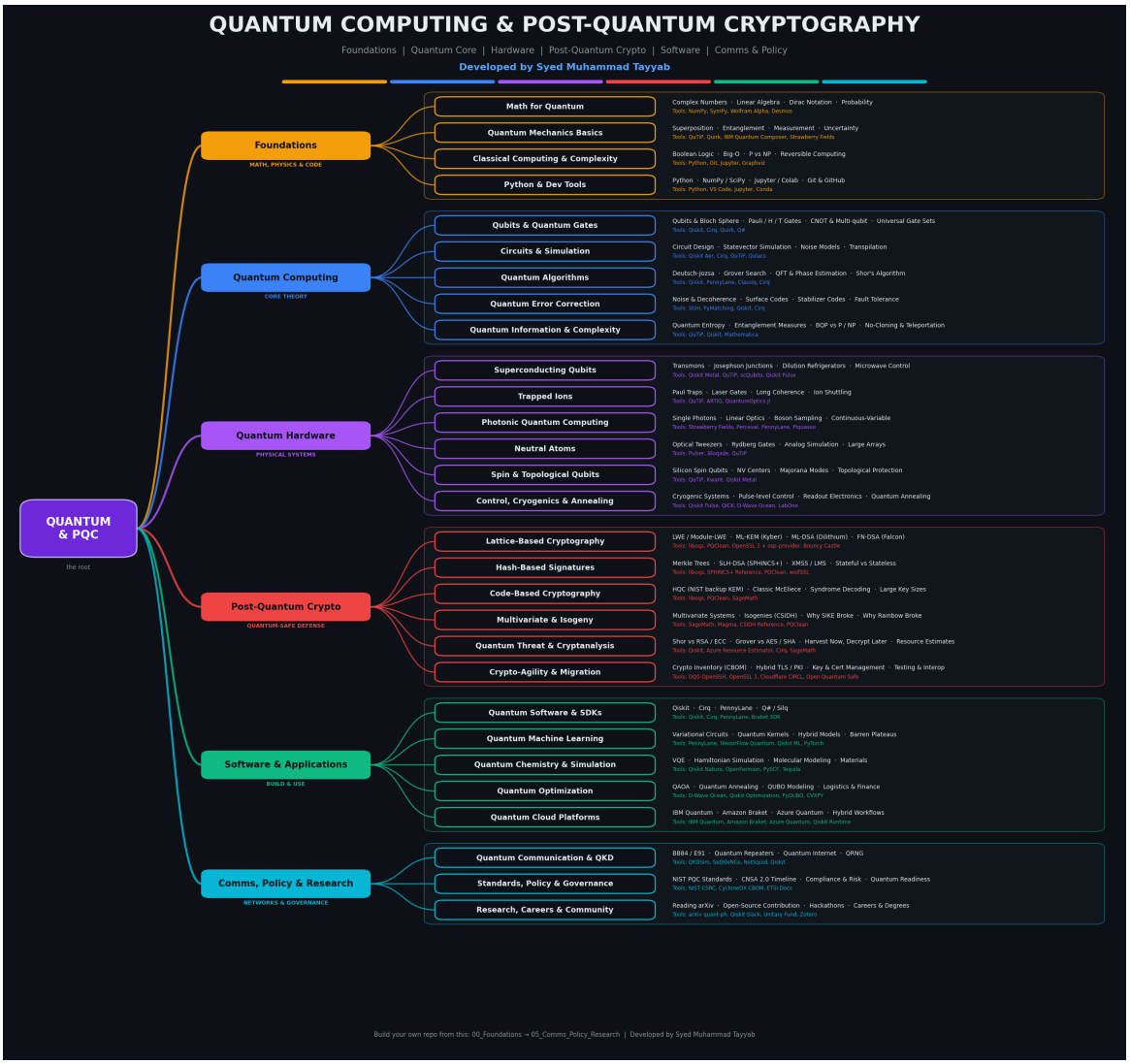
Educational reference. Follow official NIST guidance before real-world deployment.

Developed by Syed Muhammad Tayyab

github.com/Syed-Muhammad-Tayyab

1. The Full Tree

Root, 6 branches and 29 fields with their sub-fields and tools.



Branch	Type	What it does	Fields
[ ] Foundations	Math, Physics & Code	Math, quantum mechanics and code: the base of everything	4
[ ] Quantum Computing	Core Theory	Qubits, gates, circuits, algorithms and error correction	5
[ ] Quantum Hardware	Physical Systems	The machines that actually hold and control qubits	6
[ ] Post-Quantum Crypto	Quantum-Safe Defense	Cryptography that survives quantum attacks	6
[ ] Software & Applications	Build & Use	SDKs, algorithms in practice and cloud access	5
[ ] Comms, Policy & Research	Networks & Governance	Quantum networks, standards, roadmaps and careers	3

2. Foundations

Math, Physics & Code | Math, quantum mechanics and code: the base of everything

Math for Quantum `00_Foundations/Math_For_Quantum/`

Linear algebra and probability: the language of qubits. | Level ★★☆☆☆ | ~6-8 wks

Sub-fields	Complex Numbers | Linear Algebra | Dirac Notation | Probability
Tools	NumPy, SymPy, Wolfram Alpha, Desmos
Learn	3Blue1Brown Linear Algebra, MIT 18.06, Khan Academy

Quantum Mechanics Basics `00_Foundations/Quantum_Mechanics_Basics/`

Superposition, entanglement and measurement. | Level ★★★☆☆ | ~6-8 wks

Sub-fields	Superposition | Entanglement | Measurement | Uncertainty
Tools	QuTiP, Quirk, IBM Quantum Composer, Strawberry Fields
Learn	Quantum Country, MIT 8.04, Feynman Lectures

Classical Computing & Complexity `00_Foundations/Classical_Computing_Complexity/`

Bits, algorithms and what hard really means. | Level ★★☆☆☆ | ~4-6 wks

Sub-fields	Boolean Logic | Big-O | P vs NP | Reversible Computing
Tools	Python, Git, Jupyter, Graphviz
Learn	CS50, Nand2Tetris, MIT 6.006

Python & Dev Tools `00_Foundations/Python_Dev_Tools/`

Code, notebooks and version control for every experiment. | Level ★☆☆☆☆ | ~3-4 wks

Sub-fields	Python | NumPy / SciPy | Jupyter / Colab | Git & GitHub
Tools	Python, VS Code, Jupyter, Conda, Git
Learn	CS50P, Python for Everybody

3. Quantum Computing

Core Theory | Qubits, gates, circuits, algorithms and error correction

Qubits & Quantum Gates `01_Quantum_Computing/Qubits_Gates/`

The building blocks: qubits, the Bloch sphere, H, X and CNOT. | Level ★★★☆☆ | ~4-5 wks

Sub-fields	Qubits & Bloch Sphere | Pauli / H / T Gates | CNOT & Multi-qubit | Universal Gate Sets
Tools	Qiskit, Cirq, Quirk, Q#
Learn	IBM Quantum Learning, Qiskit Textbook

Circuits & Simulation `01_Quantum_Computing/Circuits_Simulation/`

Design, run and debug quantum circuits on simulators. | Level ★★★☆☆ | ~3-4 wks

Sub-fields	Circuit Design | Statevector Simulation | Noise Models | Transpilation
Tools	Qiskit Aer, Cirq, QuTiP, Qulacs, Stim
Learn	Qiskit Developer Certification

Quantum Algorithms `01_Quantum_Computing/Quantum_Algorithms/`

Where quantum beats classical: Shor, Grover and more. | Level ★★★★★ | ~8-10 wks

Sub-fields	Deutsch-Jozsa | Grover Search | QFT & Phase Estimation | Shor's Algorithm
Tools	Qiskit, PennyLane, Classiq, Cirq
Learn	Nielsen & Chuang, Qiskit Global Summer School

Quantum Error Correction `01_Quantum_Computing/Quantum_Error_Correction/`

Protect fragile qubits from noise. | Level ★★★★★ | ~8-10 wks

Sub-fields	Noise & Decoherence | Surface Codes | Stabilizer Codes | Fault Tolerance
Tools	Stim, PyMatching, Qiskit, Cirq
Learn	Preskill Lecture Notes, Error Correction Zoo

Quantum Information & Complexity `01_Quantum_Computing/Quantum_Information_Complexity/`

Entropy, entanglement measures and BQP. | Level ★★★★★ | ~6-8 wks

Sub-fields	Quantum Entropy | Entanglement Measures | BQP vs P / NP | No-Cloning & Teleportation
Tools	QuTiP, Qiskit, Mathematica
Learn	Watrous: Theory of Quantum Information, Aaronson Lectures

4. Quantum Hardware

Physical Systems | The machines that actually hold and control qubits

Superconducting Qubits `02_Quantum_Hardware/Superconducting_Qubits/`

A leading platform used by IBM and Google. | Level ★★★★★ | ~8-10 wks

Sub-fields	Transmons | Josephson Junctions | Dilution Refrigerators | Microwave Control
Tools	Qiskit Metal, QuTiP, scQubits, Qiskit Pulse
Learn	Circuit QED Lectures

Trapped Ions `02_Quantum_Hardware/Trapped_Ions/`

Highly accurate qubits held by electric fields. | Level ★★★★★ | ~6-8 wks

Sub-fields	Paul Traps | Laser Gates | Long Coherence | Ion Shuttling
Tools	QuTiP, ARTIQ, QuantumOptics.jl
Learn	Quantinuum Resources, IonQ Docs

Photonic Quantum Computing `02_Quantum_Hardware/Photonic_Quantum/`

Qubits made of light. | Level ★★★★☆ | ~5-6 wks

Sub-fields	Single Photons | Linear Optics | Boson Sampling | Continuous-Variable
Tools	Strawberry Fields, Perceval, PennyLane, Piquasso
Learn	Xanadu Codebook, Quandela Docs

Neutral Atoms `02_Quantum_Hardware/Neutral_Atoms/`

Arrays of atoms trapped by laser tweezers. | Level ★★★★★ | ~6-8 wks

Sub-fields	Optical Tweezers | Rydberg Gates | Analog Simulation | Large Arrays
Tools	Pulser, Bloqade, QuTiP
Learn	QuEra Docs, Pasqal Docs

Spin & Topological Qubits `02_Quantum_Hardware/Spin_Topological_Qubits/`

Silicon spins and exotic Majorana-based qubits. | Level ★★★★★ | ~6-8 wks

Sub-fields	Silicon Spin Qubits | NV Centers | Majorana Modes | Topological Protection
Tools	QuTiP, Kwant, Qiskit Metal
Learn	Microsoft Quantum Docs, Spin Qubit Reviews

Control, Cryogenics & Annealing `02_Quantum_Hardware/Control_Cryo_Annealing/`

Keep qubits cold, controlled and measured; plus annealers. | Level ★★★★★ | ~5-6 wks

Sub-fields	Cryogenic Systems | Pulse-level Control | Readout Electronics | Quantum Annealing
Tools	Qiskit Pulse, QICK, D-Wave Ocean, LabOne
Learn	D-Wave Learning, Qiskit Pulse Docs

5. Post-Quantum Crypto

Quantum-Safe Defense | Cryptography that survives quantum attacks

Lattice-Based Cryptography `03_Post_Quantum_Crypto/Lattice_Based/`

The main NIST picks: ML-KEM (Kyber) and ML-DSA (Dilithium). | Level ★★★★★ | ~6-8 wks

Sub-fields	LWE / Module-LWE | ML-KEM (Kyber) | ML-DSA (Dilithium) | FN-DSA (Falcon)
Tools	liboqs, PQClean, OpenSSL 3 + oqs-provider, Bouncy Castle
Learn	NIST FIPS 203 / 204, Regev LWE Paper

Hash-Based Signatures `03_Post_Quantum_Crypto/Hash_Based_Signatures/`

Signatures whose security rests only on hash functions. | Level ★★★★☆ | ~3-4 wks

Sub-fields	Merkle Trees | SLH-DSA (SPHINCS+) | XMSS / LMS | Stateful vs Stateless
Tools	liboqs, SPHINCS+ Reference, PQClean, wolfSSL
Learn	NIST FIPS 205, NIST SP 800-208

Code-Based Cryptography `03_Post_Quantum_Crypto/Code_Based/`

Error-correcting codes as a hard problem (HQC, McEliece). | Level ★★★★★ | ~4-5 wks

Sub-fields	HQC (NIST backup KEM) | Classic McEliece | Syndrome Decoding | Large Key Sizes
Tools	liboqs, PQClean, SageMath
Learn	NIST IR 8545, McEliece Papers

Multivariate & Isogeny `03_Post_Quantum_Crypto/Multivariate_Isogeny/`

Other families, and lessons from broken schemes. | Level ★★★★★ | ~4-5 wks

Sub-fields	Multivariate Systems | Isogenies (CSIDH) | Why SIKE Broke | Why Rainbow Broke
Tools	SageMath, Magma, CSIDH Reference, PQClean
Learn	NIST Round Reports

Quantum Threat & Cryptanalysis `03_Post_Quantum_Crypto/Quantum_Threat_Cryptanalysis/`

Which algorithms Shor and Grover really break. | Level ★★★★☆ | ~3-4 wks

Sub-fields	Shor vs RSA / ECC | Grover vs AES / SHA | Harvest Now, Decrypt Later | Resource Estimates
Tools	Qiskit, Azure Resource Estimator, Cirq, SageMath
Learn	Gidney & Ekera Paper, Mosca's Theorem

Crypto-Agility & Migration `03_Post_Quantum_Crypto/Crypto_Agility_Migration/`

Inventory, hybrid modes and migrating real systems. | Level ★★★☆☆ | ~4-5 wks

Sub-fields	Crypto Inventory (CBOM) | Hybrid TLS / PKI | Key & Cert Management | Testing & Interop
Tools	OQS-OpenSSH, OpenSSL 3, Cloudflare CIRCL, Open Quantum Safe
Learn	NIST NCCoE Migration, PQCA Resources

6. Software & Applications

Build & Use | SDKs, algorithms in practice and cloud access

Quantum Software & SDKs `04_Software_Applications/Quantum_SDKs/`

Frameworks to write, optimize and run quantum programs. | Level ★★★☆☆ | ~4-6 wks

Sub-fields	Qiskit | Cirq | PennyLane | Q# / Silq
Tools	Qiskit, Cirq, PennyLane, Braket SDK, Q#
Learn	Qiskit Developer Certification, Unitary Fund Tutorials

Quantum Machine Learning `04_Software_Applications/Quantum_ML/`

Quantum circuits meeting neural networks. | Level ★★★★★ | ~6-8 wks

Sub-fields	Variational Circuits | Quantum Kernels | Hybrid Models | Barren Plateaus
Tools	PennyLane, TensorFlow Quantum, Qiskit ML, PyTorch
Learn	PennyLane Codebook, Xanadu QML Course

Quantum Chemistry & Simulation `04_Software_Applications/Quantum_Chemistry/`

Simulate molecules and materials the way nature does. | Level ★★★★★ | ~6-8 wks

Sub-fields	VQE | Hamiltonian Simulation | Molecular Modeling | Materials
Tools	Qiskit Nature, OpenFermion, PySCF, Tequila
Learn	Qiskit Nature Docs, OpenFermion Tutorials

Quantum Optimization `04_Software_Applications/Quantum_Optimization/`

Search huge solution spaces with QAOA and annealing. | Level ★★★★☆ | ~4-5 wks

Sub-fields	QAOA | Quantum Annealing | QUBO Modeling | Logistics & Finance
Tools	D-Wave Ocean, Qiskit Optimization, PyQUBO, CVXPY
Learn	D-Wave Learning, Qiskit Optimization Docs

Quantum Cloud Platforms `04_Software_Applications/Quantum_Cloud/`

Run real hardware from your laptop. | Level ★★★☆☆ | ~2-3 wks

Sub-fields	IBM Quantum | Amazon Braket | Azure Quantum | Hybrid Workflows
Tools	IBM Quantum, Amazon Braket, Azure Quantum, Qiskit Runtime
Learn	IBM Quantum Learning, AWS Braket Workshops

7. Comms, Policy & Research

Networks & Governance | Quantum networks, standards, roadmaps and careers

Quantum Communication & QKD `05_Comms_Policy_Research/Quantum_Communication_QKD/`

Keys protected by the laws of physics. | Level ★★★★☆ | ~4-5 wks

Sub-fields	BB84 / E91 | Quantum Repeaters | Quantum Internet | QRNG
Tools	QKDSim, SeQUeNCe, NetSquid, Qiskit
Learn	Quantum Internet Alliance, ETSI QKD Standards

Standards, Policy & Governance `05_Comms_Policy_Research/Standards_Policy_Governance/`

NIST, NSA CNSA 2.0, ETSI and national roadmaps. | Level ★★☆☆☆ | ~3-4 wks

Sub-fields	NIST PQC Standards | CNSA 2.0 Timeline | Compliance & Risk | Quantum Readiness
Tools	NIST CSRC, CycloneDX CBOM, ETSI Docs
Learn	CISSP, ISO 27001, PQCC Roadmap

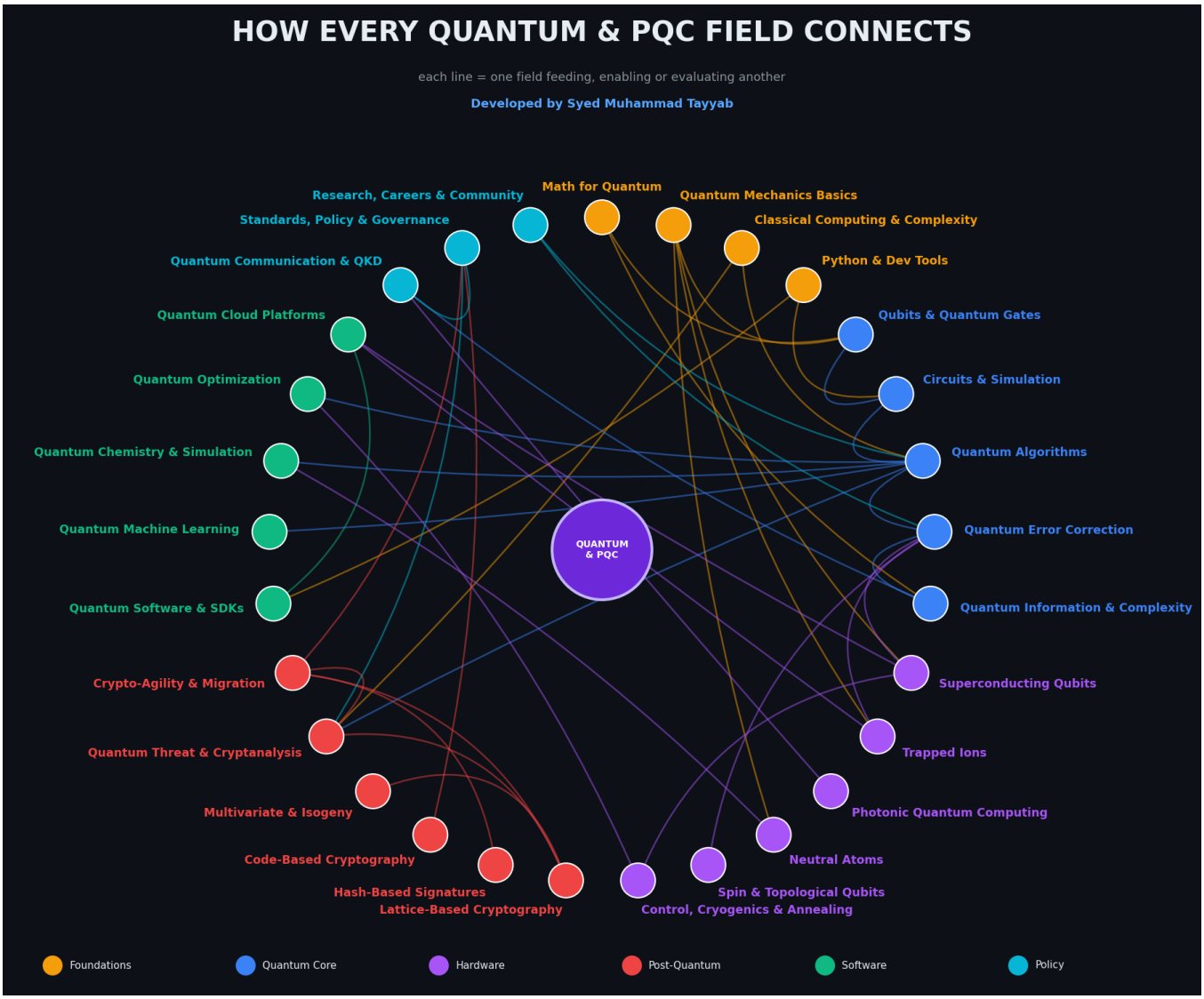
Research, Careers & Community `05_Comms_Policy_Research/Research_Careers_Community/`

Read papers, join communities, build a quantum career. | Level ★★★☆☆ | ~ongoing wks

Sub-fields	Reading arXiv | Open-Source Contribution | Hackathons | Careers & Degrees
Tools	arXiv quant-ph, Qiskit Slack, Unitary Fund, Zotero
Learn	Qiskit Advocate, Quantum Open Source Foundation

8. How Everything Is Linked

Each line = one field feeding, protecting or testing another.



8.1 Link Table

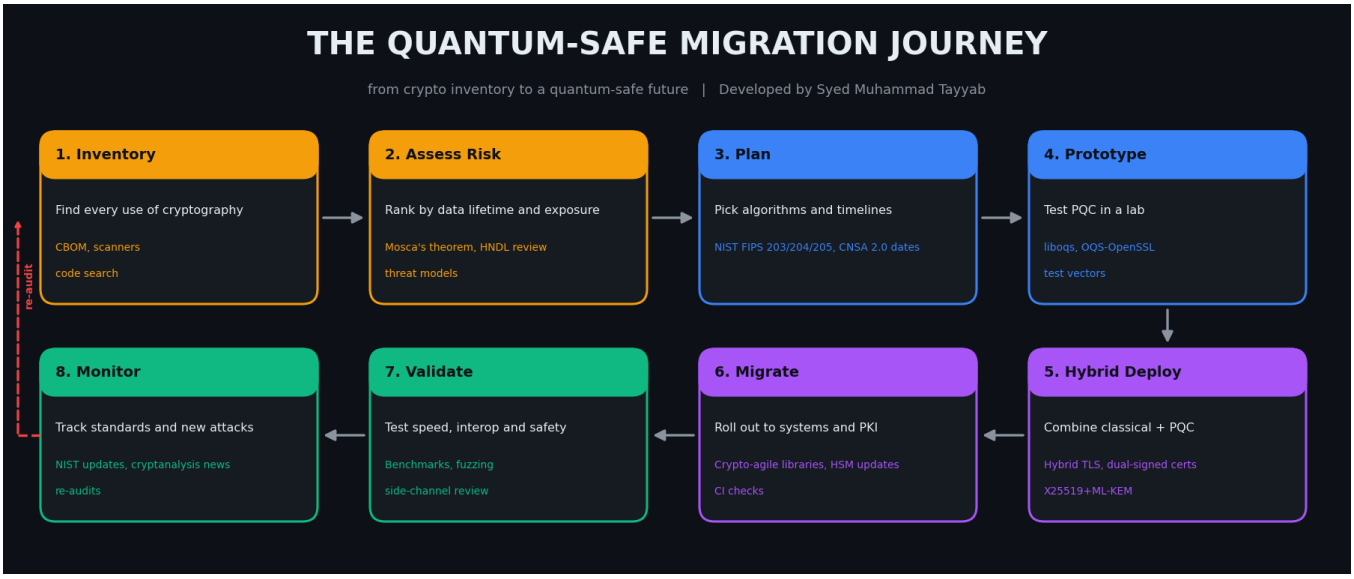
From		To	Why they connect
Math for Quantum	->	Qubits & Quantum Gates	linear algebra describes qubit states
Quantum Mechanics Basics	->	Qubits & Quantum Gates	superposition & entanglement become qubits
Math for Quantum	->	Quantum Information & Complexity	density matrices & entropy
Classical Computing & Complexity	->	Quantum Algorithms	complexity shows where quantum helps
Classical Computing & Complexity	->	Quantum Threat & Cryptanalysis	hardness assumptions behind crypto
Python & Dev Tools	->	Quantum Software & SDKs	quantum programs are Python programs
Python & Dev Tools	->	Circuits & Simulation	build & simulate circuits
Qubits & Quantum Gates	->	Circuits & Simulation	gates compose into circuits
Circuits & Simulation	->	Quantum Algorithms	algorithms are circuits
Quantum Algorithms	->	Quantum Threat & Cryptanalysis	Shor & Grover set the threat
Quantum Algorithms	->	Quantum Machine Learning	variational & kernel methods
Quantum Algorithms	->	Quantum Chemistry & Simulation	VQE & Hamiltonian simulation
Quantum Algorithms	->	Quantum Optimization	QAOA & amplitude amplification
Quantum Error Correction	->	Quantum Algorithms	fault tolerance enables big algorithms
Quantum Information & Complexity	->	Quantum Error Correction	codes built on information theory
Quantum Information & Complexity	->	Quantum Communication & QKD	entanglement & no-cloning
Quantum Mechanics Basics	->	Superconducting Qubits	circuit QED physics
Quantum Mechanics Basics	->	Trapped Ions	atomic physics
Quantum Mechanics Basics	->	Neutral Atoms	Rydberg physics
Superconducting Qubits	->	Quantum Error Correction	surface-code experiments
Trapped Ions	->	Quantum Error Correction	high-fidelity gates
Photonic Quantum Computing	->	Quantum Communication & QKD	photons carry keys
Neutral Atoms	->	Quantum Chemistry & Simulation	analog simulation
Spin & Topological Qubits	->	Quantum Error Correction	topological protection
Control, Cryogenics & Annealing	->	Superconducting Qubits	cryogenics & pulses
Control, Cryogenics & Annealing	->	Quantum Optimization	quantum annealers
Superconducting Qubits	->	Quantum Cloud Platforms	hardware via the cloud
Trapped Ions	->	Quantum Cloud Platforms	hardware via the cloud
Quantum Software & SDKs	->	Quantum Cloud Platforms	submit jobs to real devices
Quantum Threat & Cryptanalysis	->	Lattice-Based Cryptography	motivation for new hard problems
Quantum Threat & Cryptanalysis	->	Crypto-Agility & Migration	harvest-now-decrypt-later urgency
Lattice-Based Cryptography	->	Crypto-Agility & Migration	ML-KEM & ML-DSA deployment
Hash-Based Signatures	->	Crypto-Agility & Migration	stateful & stateless signatures

From		To	Why they connect
Code-Based Cryptography	->	Standards, Policy & Governance	backup NIST standard
Multivariate & Isogeny	->	Lattice-Based Cryptography	lessons from broken schemes
Crypto-Agility & Migration	->	Standards, Policy & Governance	migration timelines
Quantum Communication & QKD	->	Standards, Policy & Governance	QKD vs PQC guidance
Standards, Policy & Governance	->	Quantum Threat & Cryptanalysis	risk-based planning
Research, Careers & Community	->	Quantum Error Correction	fast-moving research
Research, Careers & Community	->	Quantum Algorithms	new algorithms & resource estimates

9. The Quantum-Safe Migration Journey

From crypto inventory to a quantum-safe future: how the fields come together.

Step	What happens	Tools & good practice
1. Inventory	Find every use of cryptography	CBOM, scanners, code search
2. Assess Risk	Rank by data lifetime and exposure	Mosca's theorem, HNDL review, threat models
3. Plan	Pick algorithms and timelines	NIST FIPS 203/204/205, CNSA 2.0 dates
4. Prototype	Test PQC in a lab	liboqs, OQS-OpenSSL, test vectors
5. Hybrid Deploy	Combine classical + PQC	Hybrid TLS, dual-signed certs, X25519+ML-KEM
6. Migrate	Roll out to systems and PKI	Crypto-agile libraries, HSM updates, CI checks
7. Validate	Test speed, interop and safety	Benchmarks, fuzzing, side-channel review
8. Monitor	Track standards and new attacks	NIST updates, cryptanalysis news, re-audits



Types of Quantum Work

Foundations (Amber)	Math, quantum mechanics and Python: the base of everything.
Quantum Computing (Blue)	Qubits, gates, algorithms and error correction.
Quantum Hardware (Purple)	Superconducting, ion, photonic and atom-based machines.
Post-Quantum Crypto (Red)	Algorithms that stay secure against quantum attacks.
Software & Apps (Green)	SDKs, chemistry, optimization, QML and cloud access.
Comms & Policy (Cyan)	Quantum networks, standards, roadmaps and research.

10. Folder Structure (your study repo)

Which field lives in which folder. Each leaf folder holds notes/, labs/, tools/ and cheatsheets/.

```
Quantum-PQC-Guide/
├── 00_Foundations/   [Math, Physics & Code]
│   ├── Math_For_Quantum/
│   ├── Quantum_Mechanics_Basics/
│   ├── Classical_Computing_Complexity/
│   └── Python_Dev_Tools/
├── 01_Quantum_Computing/   [Core Theory]
│   ├── Qubits_Gates/
│   ├── Circuits_Simulation/
│   ├── Quantum_Algorithms/
│   ├── Quantum_Error_Correction/
│   └── Quantum_Information_Complexity/
├── 02_Quantum_Hardware/   [Physical Systems]
│   ├── Superconducting_Qubits/
│   ├── Trapped_Ions/
│   ├── Photonic_Quantum/
│   ├── Neutral_Atoms/
│   ├── Spin_Topological_Qubits/
│   └── Control_Cryo_Annealing/
├── 03_Post_Quantum_Crypto/   [Quantum-Safe Defense]
│   ├── Lattice_Based/
│   ├── Hash_Based_Signatures/
│   ├── Code_Based/
│   ├── Multivariate_Isogeny/
│   ├── Quantum_Threat_Cryptanalysis/
│   └── Crypto_Agility_Migration/
├── 04_Software_Applications/   [Build & Use]
│   ├── Quantum_SDKs/
│   ├── Quantum_ML/
│   ├── Quantum_Chemistry/
│   ├── Quantum_Optimization/
│   └── Quantum_Cloud/
├── 05_Comms_Policy_Research/   [Networks & Governance]
│   ├── Quantum_Communication_QKD/
│   ├── Standards_Policy_Governance/
│   └── Research_Careers_Community/
├── 06_Projects/   # Circuits, PQC demos and benchmarks
├── 07_Labs_Notebooks/   # Qiskit, Cirq and liboqs notebooks
├── 08_Papers_Notes/   # Paper summaries (Shor, Grover, NIST PQC)
└── 09_Resources/   # Books, courses, standards, cheat sheets
```

11. Learning & Certification Roadmap

Pick a track after the foundations. Most people start with the Qiskit Textbook or IBM Quantum Learning.

Track	Certifications
Beginner	IBM Quantum Learning, Qiskit Textbook, Quantum Country, edX Quantum Mechanics
Quantum Developer	Qiskit Developer Certification, PennyLane Codebook, Cirq Tutorials, Azure Quantum Docs
PQC & Security	NIST PQC Resources, Open Quantum Safe (liboqs), ETSI Quantum-Safe Docs, CISSP/CISM + PQC reading
Research	MIT Quantum Courses, Nielsen & Chuang, Preskill Notes, arXiv quant-ph

Suggested order

1	**Foundations**	Linear algebra, quantum mechanics, Python
2	**Quantum Core**	Qubits, circuits, algorithms, error correction
3	**Pick a Path**	Quantum developer, hardware or PQC security
4	**Build**	Qiskit labs, liboqs demos, benchmarks
5	**Ship & Share**	GitHub projects, write-ups, migration plans

Portfolio Project Ideas

Level	Project	Skills
Beginner	Bell-state circuit and CHSH test on a simulator	Qubits, entanglement, Qiskit
Beginner	Quantum teleportation circuit	Gates, measurement, circuits
Intermediate	Grover search on a small puzzle	Algorithms, amplitude amplification
Intermediate	Benchmark ML-KEM vs X25519 with liboqs	PQC, performance, C / Python
Advanced	Hybrid TLS handshake lab with OQS-OpenSSL	PQC migration, TLS, PKI
Advanced	Crypto inventory (CBOM) scanner for a repository	Crypto-agility, static analysis